

ABHIRAM JAMPANI

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EDUCATION

Indian Institute of Information Technology, Lucknow

Uttar Pradesh

Bachelor of Technology - Computer Science **CGPA: 8.1**

Nov. 2022 - Present

Coursework: Object-Oriented Programming, Data Structures, Operating System, Database Management System

TECHNICAL SKILLS

Programming Languages: C++, Java, Python, JavaScript, TypeScript, SQL, Golang, HTML/CSS, CUDA

Web & Databases: React.js, Next.js, Node.js, Express.js, FastAPI, MongoDB, MySQL, PostgreSQL, Redis, REST APIs

Machine Learning & AI: TensorFlow, Scikit-learn, OCR (Google Vision, Tesseract), OR-Tools

Developer Tools & DevOps: Git, GitHub, Linux, Docker, Google Cloud (GCP), AWS, CI/CD

AI Tools: Cursor, GitHub Copilot, n8n

WORK EXPERIENCE

YugabyteDB

India

Software Engineer Intern

Jan 2026 – June 2026

- Engineered production-grade changes to **YugabyteDB's CDC and Logical Replication pipeline** — spanning the Virtual WAL layer, replication slot lifecycle, and pgoutput protocol — improving data streaming reliability for enterprise-scale distributed workloads.
- Overhauled the **logical replication decoding layer** to strengthen PostgreSQL-compatible CDC correctness, fixing replica identity handling, row-level publication filtering, and operation-type semantics across distributed nodes.
- Architected and developed **exclusive replication slot acquisition** via session-level advisory locks with a stable name-to-integer mapping, eliminating concurrent slot conflicts and improving CDC reliability in multi-tenant environments.

NVIDIA

India

Software Engineer Intern

July 2025 – Dec 2025

- Engineered a high-performance C++ library for NVIDIA's CUDA assembler and disassembler, exposing public APIs via a static library to enable external clients to integrate low-level binary analysis into their toolchains.
- Implemented parallel decoding using pthreads to process binary sections concurrently, achieving a **2× increase in analysis throughput** and improving scalability for large GPU binaries.
- Identified and resolved critical memory leaks and lifetime-management issues under multi-threaded execution, reducing build and integration latency by **85%** and accelerating compiler feature delivery.

PROJECTS

FarmChain 🔄 (Live)

Next.js, React, Node.js, Express.js, GraphQL, MongoDB, Solidity, Polygon, Kafka, JWT

- Built a full-stack food traceability platform using Next.js, Node.js, Express, GraphQL, and MongoDB, with QR-based product verification and JWT+OTP authentication to reduce fraud across the supply chain.
- Integrated Solidity smart contracts and IPFS on Polygon/Ethereum for immutable record anchoring and escrow payments, enabling end-to-end farm-to-consumer visibility with MetaMask/WalletConnect support.

ConnectExpress – Smart Railway Route Optimizer 🔄

Next.js, Node.js, Express.js, MongoDB, Redis, cmake-js,

- Built a full-stack railway search platform with a Node.js/Express backend and a custom **C++ N-API addon** that computes direct and multi-leg journeys using a meet-in-the-middle algorithm, delivering **millisecond-scale route synthesis** across thousands of train combinations.
- Designed a **MongoDB** schema for trains and stations paired with a **Redis** cache-aside layer, handling day rollovers, layover constraints, and concurrent user searches with consistent low latency and graceful degradation.

COMPETITIVE PROGRAMMING

- Rated **2139 (Guardian)** on [LeetCode](#), **1554 (Specialist)** on [Codeforces](#), and **2010 (5 Star)** on [CodeChef](#).

OPEN SOURCE & HACKATHONS

- Open Source Contributor to [LLVM](#) and [BitcoinFuzz](#), contributing production-grade code to compiler infrastructure and Bitcoin cryptography fuzzing.
- Winner** — Hack-O-Fiesta V4.0, IIIT Lucknow.

POSITIONS OF RESPONSIBILITY

- Overall Coordinator** for Equinox (Techno-Cultural Fest), IIIT Lucknow - Managed **100+ events**, led **30+ clubs**